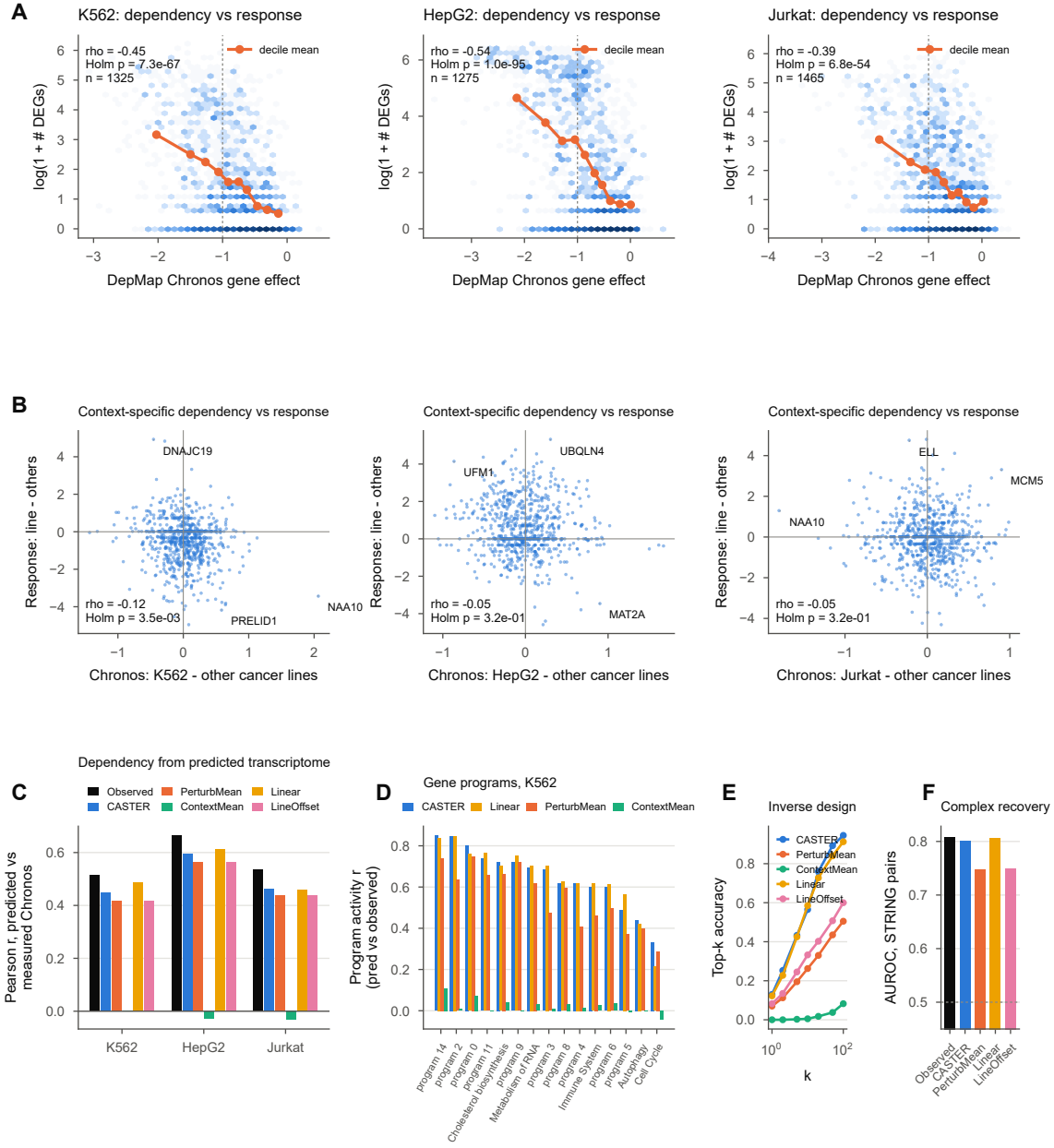

A
K562: dependency vs response
rho = -0.45
Holm p = 7.3e-67
n = 1325
decile mean
log(1 + # DEGs)
DepMap Chronos gene effect
HepG2: dependency vs response
rho = -0.54
Holm p = 1.0e-95
n = 1275
decile mean
log(1 + # DEGs)
DepMap Chronos gene effect
Jurkat: dependency vs response
rho = -0.39
Holm p = 6.8e-54
n = 1465
decile mean
log(1 + # DEGs)
DepMap Chronos gene effect
B
Context-specific dependency vs response
DNAJC19
rho = -0.12
Holm p = 3.5e-03
PRELID1
NAA10
Response: line - others
Chronos: K562 - other cancer lines
Context-specific dependency vs response
UFM1
UBQLN4
rho = -0.05
Holm p = 3.2e-01
MAT2A
Response: line - others
Chronos: HepG2 - other cancer lines
Context-specific dependency vs response
ELL
MCM5
NAA10
rho = -0.05
Holm p = 3.2e-01
Response: line - others
Chronos: Jurkat - other cancer lines
C
Dependency from predicted transcriptome
Observed
CASTER
PerturbMean
ContextMean
Linear
LineOffset
Pearson r, predicted vs measured Chronos
K562
HepG2
Jurkat
D
Gene programs, K562
CASTER
Linear
PerturbMean
ContextMean
Program activity r (pred vs observed)
program 14
program 2
program 0
program 11
Cholesterol biosynthesis
Metabolism of RNA
program 9
program 3
program 8
program 4
Immune System
program 6
program 5
Autophagy
Cell Cycle
E
Inverse design
CASTER
PerturbMean
ContextMean
Linear
LineOffset
Top-k accuracy
k
F
Complex recovery
AUROC, STRING pairs
Observed
CASTER
PerturbMean
Linear
LineOffset